PDA College Of Engineering

Department Of Computer Science and Engineering

SYNOPSIS OF THE PROPOSED MAJORPROJECT (22CS65)

ON

Talk-to-Data: A Voice-Controlled System to Search Databases

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Department Vision Mission Statement

Vision

To become a premier department in Computer education, research and to prepare highly competent IT professionals to serve industry and society at local and global levels.

Mission

- To impart high quality professional education to become a leader in Computer Science and Engineering.
- To achieve excellence in research for contributing to the development of the society.
- To inculcate professional and ethical behaviour to serve the industry.

ABSTRACT

In today's data-driven world, organizations depend on databases to store and manage large amounts of information. However, accessing and analysing this data often requires knowledge of Structured Query Language (SQL). This can create challenges for non-technical users like managers, healthcare workers, or administrative staff.

This project, "Talk-to-Data: A Noise-Resilient Natural Language Interface for Relational Databases," suggests a system that lets users query databases using simple spoken English. The system translates spoken input into SQL queries and retrieves the corresponding data from a relational database. The main idea is to combine speech recognition, natural language understanding, and database query creation into a seamless pipeline, making database interaction easier.

One major challenge with voice-based systems is environmental noise. In real settings, like offices, hospitals, or public spaces, background noise can reduce the accuracy of speech recognition. Misunderstood words can lead to incorrect SQL queries or failed database operations. To tackle this issue, the proposed system includes an audio pre-processing stage that removes background noise before speech transcription. This step improves the signal-to-noise ratio and boosts transcription accuracy.

The main goal of this project is to show that using audio denoising techniques can greatly improve the reliability of voice-driven database interfaces. This approach could be useful in various areas, including business analytics, healthcare data retrieval, and administrative information systems, where quick and intuitive access to data is crucial.

INTRODUCTION

In todays world data is very important for companies and organizations. They use databases to store and manage a lot of information. These databases are good because they help keep the data organized and make it easy to find what you need.. To get to this data you usually need to know how to use something called Structured Query Language or SQL for short. This can be hard for people who're not good with computers or do not know how to use databases.

Normally people have to write SQL commands to get the information they want. Even simple things like finding records or counting things require an understanding of how databases work. This makes it difficult for people who need the information to make decisions, but do not know Query Languages. So companies often have to rely on people who're good with computers to get the information, which can slow things down.

With new technologies like Artificial Intelligence and Natural Language Processing things are getting easier. Now people can use language to ask databases questions instead of having to write special commands. This means that users can ask questions in English and the system will understand what they want and get the right information. This makes it easier for more people to use databases.

There is a problem with using your voice to talk to computers. Background noise, like people talking or machines making noise can make it hard for the computer to understand what you are saying. This can cause mistakes, which can be a problem when you are trying to get information from a database.

The main goal of this system is to make it easy for people to get information from databases using voice commands even when there is background noise. By making it easier for people to talk to databases this project is helping to make it easier for people to get the information they need. Data is very important for companies and organizations. This system is going to help them use data more efficiently. The database will be easier to use. People will be able to get the information they need quickly. The system will make data access more efficient, intuitive and accessible, to everyone including database users.

EXISTING AND PROPOSED SOLUTIONS

- **Traditional SQL-Based Database Systems**: Most companies use databases that need users to write codes, called SQL to get or change information. These systems work well. Are used a lot but they need people to know how databases are set up and how to write these codes. People who aren't tech experts often struggle with them which makes it hard for them to get the information they need quickly.
- **Natural Language to SQL (Text-to-SQL) Systems**: Recently some new systems have been developed that let users type what they want to know in language instead of writing special codes. These systems try to understand what the user wants and turn what they type into the right code. Users still have to type what they want and sometimes they get confused with complicated sentences or unclear language.

Talk-to-Data is a voice-driven natural language interface designed to allow users to query relational databases using simple spoken English.

Key modules of the proposed system include:

- **Voice Input Module** — The system captures the user's spoken query using a microphone.
- **Audio Preprocessing and Noise Reduction** — To improve reliability in real-world environments, the system applies audio denoising techniques before speech transcription.
- **Natural Language to SQL Translation** — The transcribed text is interpreted using a Natural Language Processing framework that understands the user's intent and converts the request into a valid SQL query based on the database schema.
- **Database Query Execution and Result Display** — The generated SQL query is executed on a relational database, and the retrieved results are displayed in a clear and user-friendly interface for the user.

PROBLEM STATEMENT

Modern organizations rely heavily on databases to store and manage large amounts of information. However, accessing this data typically requires knowledge of Structured Query Language (SQL), which creates a barrier for non-technical users who need quick access to information. Although Natural Language Processing has enabled systems that convert natural language queries into SQL, many existing solutions still require users to type their queries and may not provide a fully intuitive interaction method.

Voice-based database interfaces offer a more natural way for users to interact with data, but their performance is often affected by environmental noise. Therefore, there is a need for a reliable voice-driven database system that can handle noisy audio inputs, accurately interpret user queries, and generate correct SQL commands to retrieve the required data.

METHODOLOGY

1. **Voice Input Capture:**

   The system begins by capturing the user's spoken query through a microphone using an audio recording library. The recorded audio is stored temporarily in a digital audio format so that it can be processed by the subsequent modules in the system.

2. **Audio Preprocessing (Noise Reduction):**

   Since real-world environments often contain background noise, the recorded audio is passed through a noise reduction module. This stage applies filtering techniques to remove unwanted sounds.

3. **Speech-to-Text Conversion:**

   After preprocessing, the cleaned audio is processed using an Automatic Speech Recognition (ASR) model that converts the spoken input into a text transcript.

4. **Natural Language Understanding:**

   The generated text is then analyzed using Natural Language Processing techniques to understand the intent of the user's query. The system identifies key elements such as requested data fields, conditions, and database entities required to construct the query.

5. **SQL Query Generation:**

   Based on the interpreted intent and the structure of the database schema, the system generates a syntactically correct SQL query. This query represents the structured command required to retrieve the relevant data from the database.

6. **Database Query Execution:**

   The generated SQL command is executed on a relational database management system. The database processes the query and retrieves the corresponding data that matches the requested conditions.

7. **Result Visualization and Output:**

   Finally, the retrieved data is displayed to the user through a simple and user-friendly interface. The results are presented in a structured format such as tables, making it easier for users to interpret the information obtained from the database.

SYSTEM REQUIREMENTS

1. Hardware Requirements

- **Processor:** Intel i5 or higher (recommended for running AI models)
- **RAM:** Minimum 8 GB (16 GB recommended for smoother processing)
- **Storage**: At least 20 GB free disk space for datasets, models, and project files
- **Internet Connection:** Stable internet connection for installing dependencies and accessing AI models/APIs
- **Input Devices:** Microphone for capturing voice queries, keyboard and mouse
- **Output Devices:** Monitor or display screen

2. Software Requirements

- **Operating System:** Windows / macOS / Linux (Ubuntu recommended)
- **Programming Language:** Python
- **Frontend Framework:** Streamlit (for web interface)
- **Backend Libraries:** Python libraries for audio processing and database interaction
- **Database:** SQLite / MySQL (for storing relational data)
- **AI Models:** Whisper (Speech Recognition), LLM framework for Text-to-SQL generation
- **Libraries/Tools:** Sounddevice, Noisereduce, Pandas, Vanna.ai

SYSTEM SPECIFICATIONS

Reliability: The system should accurately convert spoken queries into SQL commands and retrieve correct database results even with moderate background noise.

Performance: Speech processing, query generation, and database retrieval should occur with minimal delay to ensure smooth user interaction.

Usability: The interface should allow users to easily submit voice queries and view results without requiring knowledge of SQL or database structures.

Scalability: The system should support expanding databases and additional queries without significant performance degradation.

Security: The system should ensure safe database access and prevent unauthorized modification of stored data.

EXPECTED OUTCOME OF THE PROJECT

The proposed project is expected to develop a functional voice-based system that allows users to interact with relational databases using simple spoken English. Instead of writing complex SQL queries, users will be able to retrieve information by speaking their requests, making database access easier for non-technical users.

The system will be capable of converting spoken input into text, interpreting the user's intent, and automatically generating accurate SQL queries based on the database structure. These queries will then be executed on a relational database, and the retrieved results will be displayed to the user through a simple and user-friendly interface.

Another important outcome of the project is the integration of an audio preprocessing module that reduces background noise before speech recognition. This feature is expected to improve the accuracy of speech-to-text conversion in noisy environments and make the system more reliable for real-world usage.

Overall, the project aims to demonstrate that combining speech recognition, natural language processing, and noise reduction techniques can create an efficient and intuitive system for accessing database information through voice commands.